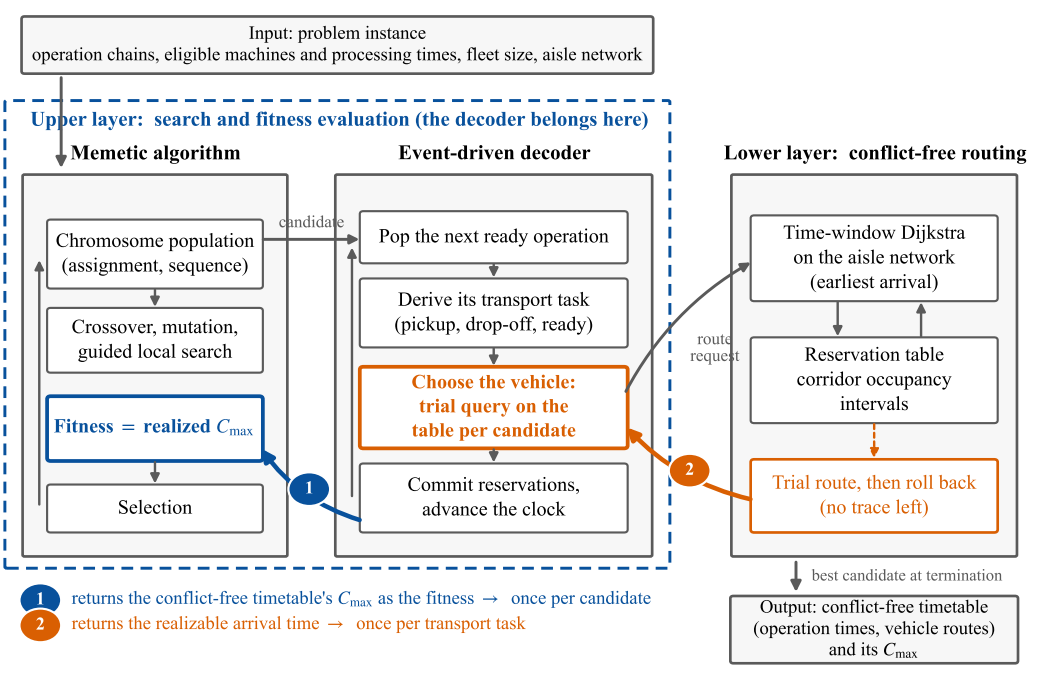
Input: problem instance
operation chains, eligible machines and processing times, fleet size, aisle network
Upper layer: search and fitness evaluation (the decoder belongs here)
Memetic algorithm
Event-driven decoder
Lower layer: conflict-free routing
Chromosome population (assignment, sequence)
Crossover, mutation, guided local search
Fitness = realized $C_{\max}$
Selection
candidate
Pop the next ready operation
Derive its transport task (pickup, drop-off, ready)
Choose the vehicle: trial query on the table per candidate
Commit reservations, advance the clock
route request
Time-window Dijkstra on the aisle network (earliest arrival)
Reservation table corridor occupancy intervals
Trial route, then roll back (no trace left)
best candidate at termination
Output: conflict-free timetable (operation times, vehicle routes) and its $C_{\max}$
1
2
1 returns the conflict-free timetable's $C_{\max}$ as the fitness → once per candidate
2 returns the realizable arrival time → once per transport task